Formula sheet for Quiz 8

Boxes

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

An anchor is a fixed box (x_a, y_a, w_a, h_a). The network predicts offsets (t_x, t_y, t_w, t_h) and the box is recovered as

$$x = x_a + t_x w_a, \qquad y = y_a + t_y h_a, \qquad w = w_a e^{t_w}, \qquad h = h_a e^{t_h}$$

Anchors per location = (number of scales) × (number of aspect ratios).

The YOLO output tensor

For an $S \times S$ grid, B boxes per cell and C classes, with each box carrying 4 coordinates and 1 objectness score:

$$\text{depth per cell} = B \times 5 + C, \qquad \text{output} = S \times S \times (B \times 5 + C)$$

Convolution output size

$$H_{\text{out}} = \left\lfloor \frac{H_{\text{in}} + 2p - k}{s} \right\rfloor + 1$$

Transposed convolution, which goes the other way:

$$H_{\text{out}} = (H_{\text{in}} - 1)\, s - 2p + k$$

Segmentation losses

For predicted region P and true region G:

$$\text{Dice} = \frac{2\,|P \cap G|}{|P| + |G|}, \qquad \mathcal{L}_{\text{Dice}} = 1 - \text{Dice}$$

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$$

Pixel accuracy = correct pixels / total pixels.

Patches

A square image of side S cut into patches of side P gives

$$\left(\frac{S}{P}\right)^2 \text{ patches, each of } P \times P \times 3 \text{ numbers}$$

Tokens and embeddings

`nn.Embedding(V, d)` maps $(N, T) \longrightarrow (N, T, d)$ and holds $V \times d$ parameters.

A one-hot sequence matrix $\mathbf{O}$ is $s \times n$ for s tokens over a vocabulary of n. The embedding is

$$\mathbf{E} = \mathbf{O}\,\mathbf{W}_{\mathbf{E}}, \qquad (s \times n)(n \times m) = (s \times m)$$

Reserved ids, by convention: `<pad>` = 0, `<unk>` = 1.